Hang Jung Ling, Ph.D. student

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Education

2021 – · · · · **Ph.D.**, CREATIS, INSA Lyon, France.
Thesis title: *Physics-guided neural networks for intraventricular vector flow mapping by color Doppler.*
Supervisors: Dr. Damien Garcia, Prof. Olivier Bernard, and MD-PhD Pierre-Yves Courand

2020 – 2021 **M.Sc. Medical Imaging, Signals and Systems**, Université Claude Bernard Lyon 1, France.

2018 – 2021 **M.Sc. (*Diplôme d'ingénieur*) Electrical Engineering**, INSA Lyon, France. (First Class Honours)

2016 – 2018 **Preparatory class, ASINSA**, INSA Lyon, France.

Research Publications

Journal Articles

1. **H. J. Ling**, S. Bru, J. Puig, F. Vixège, S. Mendez, F. Nicoud, P.-Y. Courand, O. Bernard, and D. Garcia, "Physics-Guided Neural Networks for Intraventricular Vector Flow Mapping," *IEEE Trans. Ultrason. Ferroelectr. Freq. Control*, pp. 1–1, 2024.
2. J. Puig, D. Friboulet, **H. J. Ling**, F. Varray, J. Porée, J. Provost, D. Garcia, and F. Millioz, "Boosting Cardiac Color Doppler Frame Rates with Deep Learning," *arXiv:2404.00067*, 2024.
3. **H. J. Ling**, O. Bernard, N. Ducros, and D. Garcia, "Phase Unwrapping of Color Doppler Echocardiography using Deep Learning," *IEEE Trans. Ultrason. Ferroelectr. Freq. Control*, vol. 70, no. 8, pp. 810–820, Aug. 2023.

Conference Proceedings

1. **H. J. Ling**, N. Painchaud, P.-Y. Courand, P.-M. Jodoin, D. Garcia, and O. Bernard, "Extraction of Volumetric Indices from Echocardiography: Which Deep Learning Solution for Clinical Use?" In *Functional Imaging and Modeling of the Heart*, 2023, pp. 245–254.

Presentations

Oral

Sep. 2023 Intraventricular Vector Flow Imaging using Physics-Informed Deep Learning
IEEE International Ultrasonics Symposium, Montreal, Canada

June 2023 Intraventricular Vector Flow Imaging using Physics-Informed Deep Learning
Artimino Conference on Medical Ultrasound Technology, Artimino, Italy

Extraction of volumetric indices from echocardiography—which deep learning solution for clinical use?
Functional Imaging and Modeling of the Heart, Lyon, France

Oct. 2022 Dealiasing of color Doppler echocardiography using deep learning
IEEE International Ultrasonics Symposium, Venice, Italy

Presentations (continued)

Poster

Mar. 2024 — Physics-Guided Neural Networks for Intraventricular Vector Flow Mapping
Colloque Français d'Intelligence Artificielle en Imagerie Biomédicale, Grenoble, France

Oct. 2022 — Reaching intra-observer variability in 2-D echocardiographic image segmentation with a simple U-Net architecture
IEEE International Ultrasonics Symposium, Venice, Italy

Skills

Languages — Chinese, English, French, and Malay.

Coding — C++, HTML/CSS, Java, Matlab, Python, and PyTorch

Miscellaneous Experience

Reviews

2023 – · · · · — **Reviewer**, IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control.

Service and Volunteer

June 2023 — **Conference organizing committee**, Functional Imaging and Modeling of the Heart.

Apr. 2023 — **Summer school organizing committee & Hands-on sessions' moderator**, Deep Learning for Medical Imaging summer school.

Honors and Awards

2021 — 3-year scholarship for doctoral study
MEGA doctoral school (ED 162), INSA Lyon, France

2015 — 6-year scholarship for bachelor's and master's degree studies abroad
Public Service Department or Jabatan Perkhidmatan Awam (JPA), Malaysia